Verbal Overshadowing in Taste Recognition Ability

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BACKGROUND

Verbal overshadowing (VO) refers to impaired recognition of a stimulus after it has been verbally described. This effect was first identified in face recognition [1] and subsequently extended to a wide range of visual stimuli, including abstract figures and maps. VO has also been observed outside the visual domain, including olfaction [2]. A common feature of these stimuli is their resistance to precise verbal description. Gustation is even more lexically impoverished than vision in both English [3] and Chinese [4], yet VO in taste recognition remains underexplored. Early work [5] suggested a VO effect among wine drinkers but not among non-drinkers or wine experts, whereas Hughson and Boakes (2009) [6] reported the opposite pattern, a verbal facilitation effect. These inconsistencies may stem from differing criteria for defining drinking frequency and expertise. To address this gap, the present study incorporated a rigorous pre-test to ensure comparable gustatory function across participants while also assessing their verbalisation ability and confidence ratings, thereby controlling for key sources of variability that may have contributed to earlier mixed findings. The present study aimed to advance VO research by shifting from the well-studied visual domain to the far less charted interaction between gustatory memory and language.

Research Questions

Research Question 1: To what extent is gustatory recognition susceptible to verbal overshadowing?
Hypothesis 1: Based on robust evidence of verbal overshadowing in visual recognition memory [7], verbalization of a target taste was predicted to impair subsequent gustatory recognition relative to a nonverbal control condition.

Research Question 2: Does explicit awareness of verbal overshadowing attenuate its negative effect on taste recognition?
Hypothesis 2: Based on prior evidence from visual [7] and olfactory recognition [2], explicit awareness of verbal overshadowing was not expected to reduce its detrimental impact on taste recognition accuracy.

Research Question 3: Does explicit awareness of verbal overshadowing combined with active suppression of description-based memory attenuate the verbal overshadowing effect in gustatory recognition?
Hypothesis 3: Based on evidence from visual recognition [7], explicit awareness coupled with active suppression of verbal descriptions was not expected to reduce the negative effect of verbalization on taste recognition.

Results

Experiment 1: Recognition accuracy was significantly lower in the Verbal group than in the Control group ($\beta = -1.63$, $SE = 0.48$, $t = -3.41$, $p = .001$), demonstrating a robust VO effect for taste (Fig. 3).

Experiment 2: Recognition accuracy remained significantly lower than in the Control group ($\beta = -1.05$, $SE = 0.48$, $t = -2.16$, $p = .034$) (Fig. 3), indicating that awareness alone does not attenuate VO.

Experiment 3: Accuracy was significantly lower than in the Control group ($\beta = -1.26$, $SE = 0.53$, $t = -2.39$, $p = .0197$) (Fig. 3), suggesting that even active suppression does not eliminate verbal overshadowing.

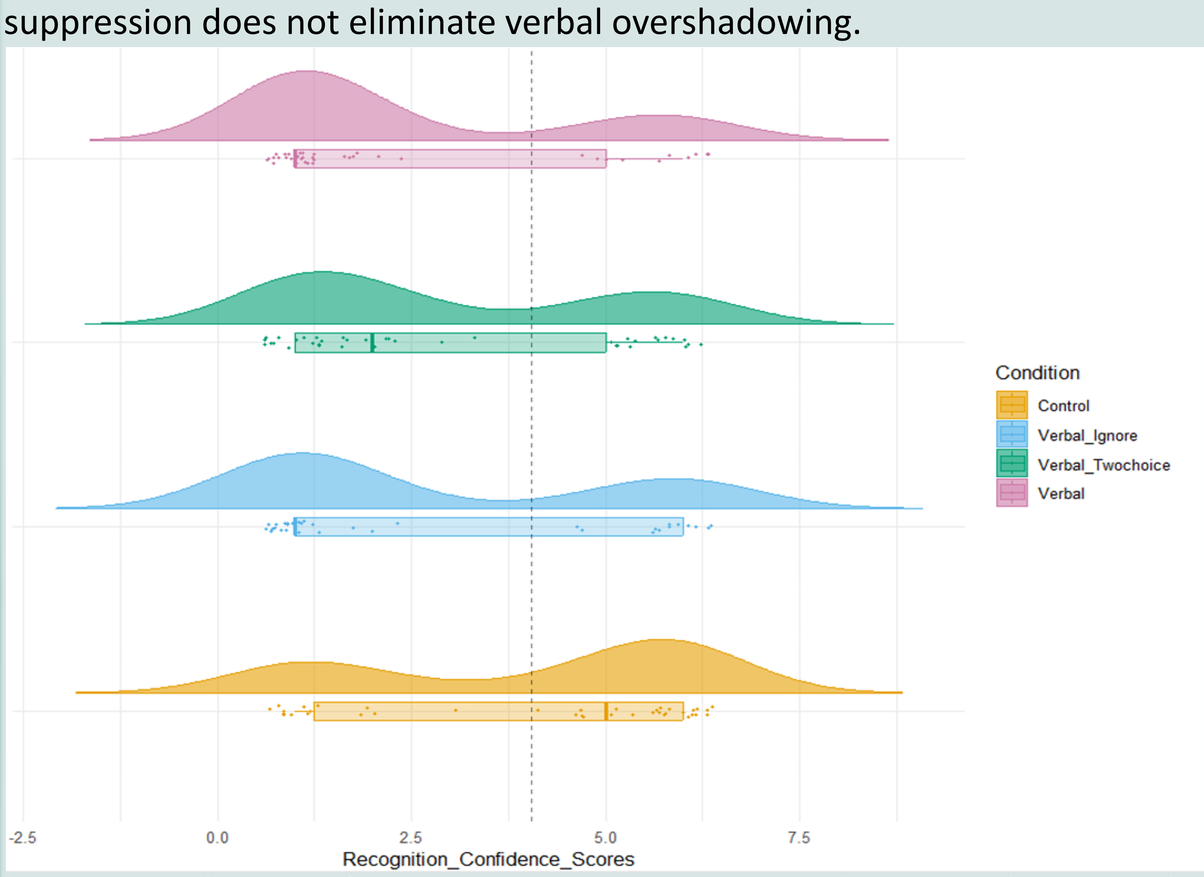

Fig. 3. Distribution of composite scores across conditions, including verbal (Experiment 1), verbal two-choice (Experiment 2), verbal ignore (Experiment 3), and control group. The dashed line marks the mean of the Control group.

METHODOLOGY

Across three experiments, participants identified target tastes from sets of four food samples (one target, three distractors; Fig. 1).

	Target	Distractors		
Set 1	Apricot	Peach	Apple	Cherry
Set 2	Kiwifruit	Pineapple	Blueberry	Grape

Fig. 1. Taste stimuli used across the three experiments.

Experiment 1 tested 40 native Chinese speakers, with 20 completing a verbal description of the target taste before recognition and 20 completing a filler task (Fig. 2).

Experiment 2 examined whether explicitly alerting participants to the possibility of VO would mitigate its effect. Twenty participants in the Verbal Two-choice group were informed that their verbal descriptions might differ from their later taste memory (Fig. 2).

Experiment 3 tested whether both awareness of VO and active suppression of description-based representations could reduce its impact. Twenty participants in the Verbal Ignore group were instructed to disregard any information derived from their descriptions and rely solely on taste memory (Fig. 2).

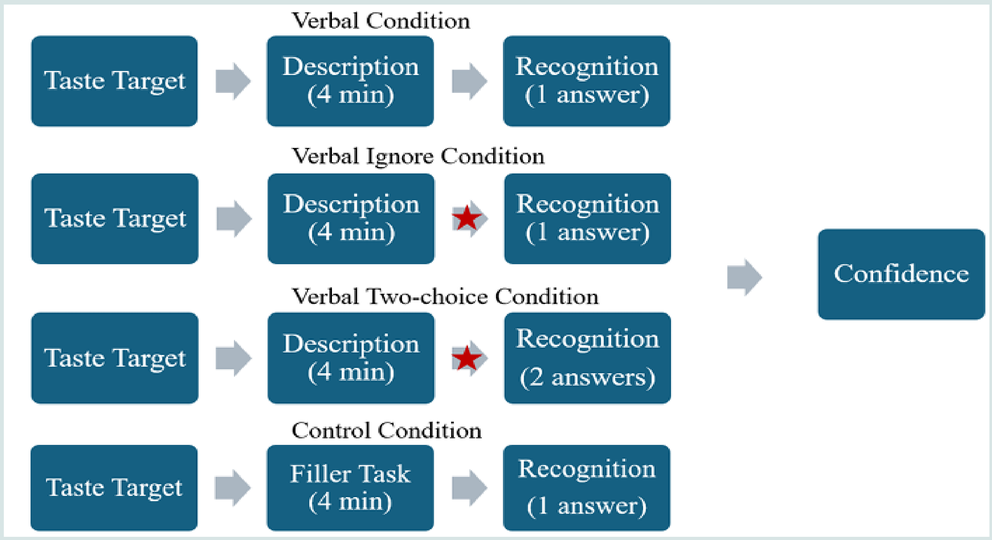

Fig. 2. Flow chart for the four conditions. The red star refers to the instruction of unveiling VOE before the recognition task.

Discussion

- √ Verbalization extends beyond visual [1,7] and olfactory [2] memory to gustatory memory, reliably impairing gustatory recognition ability.
- × Neither **increased metacognitive awareness of VO** nor **instructed suppression of verbal information** eliminates recognition impairment, consistent with findings in visual [1,7] and olfactory [2] domains.
- ★ Together, the results are discussed in relation to four candidate mechanisms, including recoding interference [1], transfer inappropriate processing shifts [8], retrieval-based interference [9], and criterion-shift accounts [10].

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